

This is an in-terminal tic-tac-toe game. Players 'X' and 'O' take turns selecting locations on the board until one player wins or there is a draw. The input must be a number between 0 and 2 describing a location on the board. After each turn, the board is updated and the current player switches.

My approach to this project was to use separate classes for the game, the player and the board.

The player class is initialized with a symbol and its turn (which is automatically set to false) and it contains the method `take_turn()`. The board class is initialized with an empty tic-tac-toe board that contains dashes in place of the symbols. This class has a method `print_board()` which iterates through each symbol in the board and then also adds lines in between to draw the board. The game class is initialized with a game board, player X and player O, the current player index, and whether the game is over. It contains a property method that returns a current player. It also has a `start_game()` method that loops through the whole game and methods that check whether someone has won or if there's been a draw. This organization of classes allowed each individual item to update individually throughout the game and also have properties and attributes that I could modify. The game just automatically runs after running the file and it keeps going until there's a draw or the game is won. To validate the input, I checked whether it was an int and whether the int was within the bounds of zero and two. If it fell outside of these bounds or was not an int then an error message would be displayed.

The data structure I used was a nested list.

Some challenges I faced were making sure that the end game condition worked properly and checking each row column and diagonal for a win. I also had to figure out how to efficiently loop through the rows and columns and diagonals within the check winner method.

Some possible improvements I could make would be to allow you to play again and keep a score between player X and player O. Another improvement would be to change colors, so that X would be one color and O would be another color.